

Zwitterion mediated anti-protein adsorption on polypropylene mesh to reduce inflammation for efficient hernia repair

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ABSTRACT

The effectiveness of polypropylene (PP) mesh is often compromised by severe inflammation. Engineering anti-inflammatory coatings has significant implications for PP mesh to repair unwanted hernias. Here, we presented a facile strategy to develop an anti-fouling coating consisting of zwitterionic poly(carboxybetaine methacrylate) (PCBMA), which could prohibit protein adsorption to endow PP mesh with anti-inflammatory efficacy. The incorporation of PCBMA coating had little impact on the raw features of PP mesh. While the modified mesh PCBMA-PP possessed noticeable hydrophilicity increase and surface charge reduction. The excellent lubricity and surface stability enabled PCBMA-PP to exhibit superior anti-fouling capacity, thus efficiently inhibiting the adsorption of proteins. *In vivo* experiments showed that incorporating the PCBMA layer could provide PP meshes with outstanding anti-inflammatory effects and tissue compatibility for repairing hernias.

1. Introduction

Abdominal wall hernia is one of the most common diseases, with an overall incidence of about 14 % [1]. The introduction of prosthetic meshes has been proven as a more effectual treatment to repair hernias [2–4]. It is estimated that over 20 million hernia repair prostheses are used annually around the world [5,6]. Polypropylene (PP) mesh is the most popular because of various merits like lightweight, excellent compliance, and high strength [7–10]. Due to inevitable foreign body reaction (FBR), many complications after surgeries are still common, including chronic pain, permanent adhesion, infection, and inflammation, causing severe harm to patients' suffering and burden [11–14]. Thereby, attenuating FBR of polymeric meshes is the key to reducing complication rates.

FBR is considered a local chronic inflammatory reaction to intracorporeal implants, likely causing implant failure [15–17]. After implantation, PP mesh can induce host inflammatory response starting from the adsorption of nonspecific proteins, followed by migration and adhesion of macrophages that release various proinflammatory cytokines, thus aggravating levels of inflammation [18,19]. To attenuate inflammation, developing anti-protein adsorption meshes can be an

effective strategy [20,21]. Constructing a hydrophilic surface on implants is a common theme for protein resistance [22–24]. Poly (ethylene glycol) (PEG) is commonly termed as the gold standard for antifouling polymers [25]. Due to its high hydrophilicity, PEG can reduce protein adsorption by decreasing access to a more hydrophobic surface onto which proteins prefer to bind [26,27]. However, PEG is susceptible to oxidation, which limits its practical use [28,29]. Hence, a new modification agent with excellent anti-fouling capability is highly desired to be the substitute for PEG.

Despite multiple anti-fouling agents appearing, few of them have been applied to resist FBR. This is mainly ascribed to the requirement for anti-FBR materials, especially stable anti-fouling abilities to prevent fouling in complex *in vivo* environments [16,30]. With equal proportions of positively and negatively charged groups, zwitterionic materials exhibit a balanced neutral charge with strong hydration via ionic solvation, thus inducing fouling resistance. [31–33]. Recently, zwitterionic polymers have attracted much attention due to their super stability and anti-fouling feature. Golabchi et al. found that anti-fouling zwitterionic coating could effectively suppress the acute inflammatory brain tissue response to implants [34]. Horne et al. reported that zwitterionic hydrogel thin film coating polydimethylsiloxane sheet enhanced anti-

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fibrotic effect and mitigated the FBR [35]. Zhou et al. stated that zwitterionic grafted PVA hydrogels could significantly prohibit cell and bacterial adhesion [36]. Despite the progress in zwitterionic surface coating, anti-fouling zwitterionic coating on the inert surface of PP meshes is still a challenge. Particularly, some limitations for widely utilized surface-initiated polymerization included harsh reaction conditions and difficulties in retaining the character of the meshes.

Mussel-inspired surface functionalization has demonstrated material-independent surface coating ability, easy implementation, biocompatibility, and most importantly, secondary modifications [37,38]. Poly(carboxybetaine methacrylate) (PCBMA) is a well known zwitterionic polymer for its stability and anti-fouling feature. However, the application of PCBMA as an anti-FBR agent and inhibiting the inflammation of PP meshes has not been verified to the best of our knowledge. In this work, we developed an anti-fouling coating that could prohibit protein adsorption to provide PP mesh with enhanced anti-inflammatory effects. The anti-fouling coating, made of zwitterionic PCBMA, was directly polymerized and grafted on the surface of PP mesh that was activated by dopamine and initiated by free radical polymerization of tertiary amine/photoinitiator. The prepared meshes were tested for physical, chemical, and mechanical properties to assess the effect of modification processes. The anti-fouling capabilities were characterized by investigating water contact angle (WCA), surface charge, hydration lubrication, and anti-protein adsorption behaviors. Additionally, the stability, cytocompatibility, and *in vitro/vivo* anti-inflammation studies were further performed to evaluate the potential application for hernia repair.

2. Materials and method

2.1. Materials

The PP mesh was supplied by Changzhou Runyuan Medical Supplies Technology, China. Dopamine hydrochloride (DA, 99 %) was provided by Sigma-Aldrich, USA. Hydrochloric acid (HCl, 36–38 %), tri (hydroxymethyl) aminomethane (Tris, ≥ 99.0 %), ethyl alcohol (≥ 99.7 %), and phosphate-buffered saline (PBS) were obtained from Sino-pharm, China. *N,N*-dimethyl ethylenediamine (DMEDA, 98 %) and carboxybetaine methacrylate (CBMA, ≥ 98 %) were purchased from Shanghai Yien Chemical Technology, China. Benzophenone (BP, 99 %) was supplied by Adamas-bet, Switzerland. Bovine serum albumin (BSA), fluorescein isothiocyanate (FITC), and cell counting kit-8 (CCK-8) were provided by Yeasen Biotechnology, China. 4', 6-diamidino-2-phenylindole (DAPI), mouse tumor necrosis factor- α (TNF- α), interleukin-6 (IL-6), and TRIzol reagent were purchased from Thermo Fisher Scientific, USA. The L929 mouse fibroblasts and RAW 264.7 macrophages were obtained from Shanghai Institute of Biochemistry and Cell Biology, China.

2.2. Fabrication of PCBMA-PP

First, DA was dissolved in Tris-HCl buffer (50 mM, pH = 8.5) at a concentration of 0.2 % (w/v). Then, PP meshes were immersed into the DA solution at 37 °C for 24 h under an aerobic environment. To eliminate useless chemicals, the mesh samples were rinsed with a mixed solution of deionized water and ethanol by ultrasonic. Finally, the resulting meshes were named PDA-PP.

For the preparation of DMPDA-PP, PDA-PP was put into DMEDA solution (3 % w/v, prepared by dissolving DMEDA in Tris-HCl buffer). The reaction continued for 12 h. Afterward, the DMPDA-PP mesh was oscillated with ultrasound to remove impurities and dried at room temperature.

The pretreatment solution was produced by dissolving BP in absolute ethanol at a concentration of 1 % (w/v). DMPDA-PP was soaked in BP solution for 30 min and dried under dark conditions to obtain the photoinitiator-loaded DMPDA-PP mesh. Next, the obtained mesh was

immersed in 0.1 M CBMA reaction solution, which was fixed in two pieces of quartz glass and exposed to UV light (365 nm) at 5 cm for 30 min, followed by a rinse in the mixture of deionized water and ethanol. After being dried totally, the prepared mesh was named PCBMA-PP.

2.3. Morphology and chemical structure

The micromorphology of each sample was observed by scanning electron microscopy (SEM, TM-3000, Hitachi, Japan). Surface chemical groups of different meshes were measured using Fourier transform infrared spectrometer (FTIR, PerkinElmer, USA). The chemical composition was analyzed with X-ray photoelectron spectroscopy (XPS, Escalab 250Xi, Thermo Fisher Scientific, USA). The basic structure image of each mesh was taken under the microscope (PXS8-T, Shanghai Cewei Optoelectronic Technology, China). Their porosities were calculated by Image J. The weight of mesh samples before and after modification was evaluated by electronic balance (TP2012, Sartorius, Germany).

2.4. Uniaxial tensile testing

The tensile testing was conducted according to the standard of ASTM D5035-11 using a YG(B)026G-500 tensile strength tester (Wenzhou Darong Textile Instrument, China). Each sample was cut into a rectangle of 25 mm $\times$ 70 mm with the longest dimension corresponding to the longitudinal/transverse direction. The distance between the upper and lower collets was set to 50 mm, and the stretching speed was 200 mm/min.

2.5. Wettability and ζ -potential

The wettability change was verified by the water contact angle (WCA) using a contact angle measuring device (OCA15EC, Data Physics, Germany). With negatively charged polystyrene beads used as tracer particles, the surface ζ -potential was determined by Zetasizer (Nano-ZS, Malvern, UK). The thickness of PCBMA coatings was investigated by spectroscopic ellipsometry (M-2000UI, J.A.Woollam, USA).

2.6. Hydration lubrication

The lubrication property was assessed using the fiber friction coefficient instrument (XCF-1A, Xinxian, China) under the mode of friction roller rotation. Before the testing, all samples were immersed in deionized water for 2 h. The rotation speed was set as 30 rpm, a preload of 0.1 cN was applied, and the decline rate of friction roller was 10 mm/min. Each sample was tested at least five times.

2.7. Protein adsorption

BSA was selected to investigate the changes in protein adsorption on meshes before and after modification. Mesh samples (1 cm $\times$ 1 cm) were incubated in 1 mg/mL BSA solution at 37 °C for 12 h. Afterward, each mesh was rinsed with PBS solution (pH = 7.4) three times and then stained with 0.1 M Na₂CO₃/NaHCO₃ buffer (pH = 9.0) containing FITC (1 mg/mL) at 4 °C for 6 h. and immersed in FITC-labeled BSA solution in the dark for 12 h. The labeled proteins on samples were observed under an inverted fluorescence microscope (Ti-S, Hitachi, Japan).

2.8. Stability of PCBMA-PP

The assay was first based on the static degradation samples to explore the stability of the modified mesh (PCBMA-PP). Each sample was put into the PBS at 37 °C for 7 days. The change in mesh morphologies during degradation was determined by SEM after incubation of 1, 3, 5, and 7 days.

The dynamic stability was evaluated according to the standard of FZ-T 70006-2004. Mesh samples were cut into 50 mm $\times$ 20 mm in the

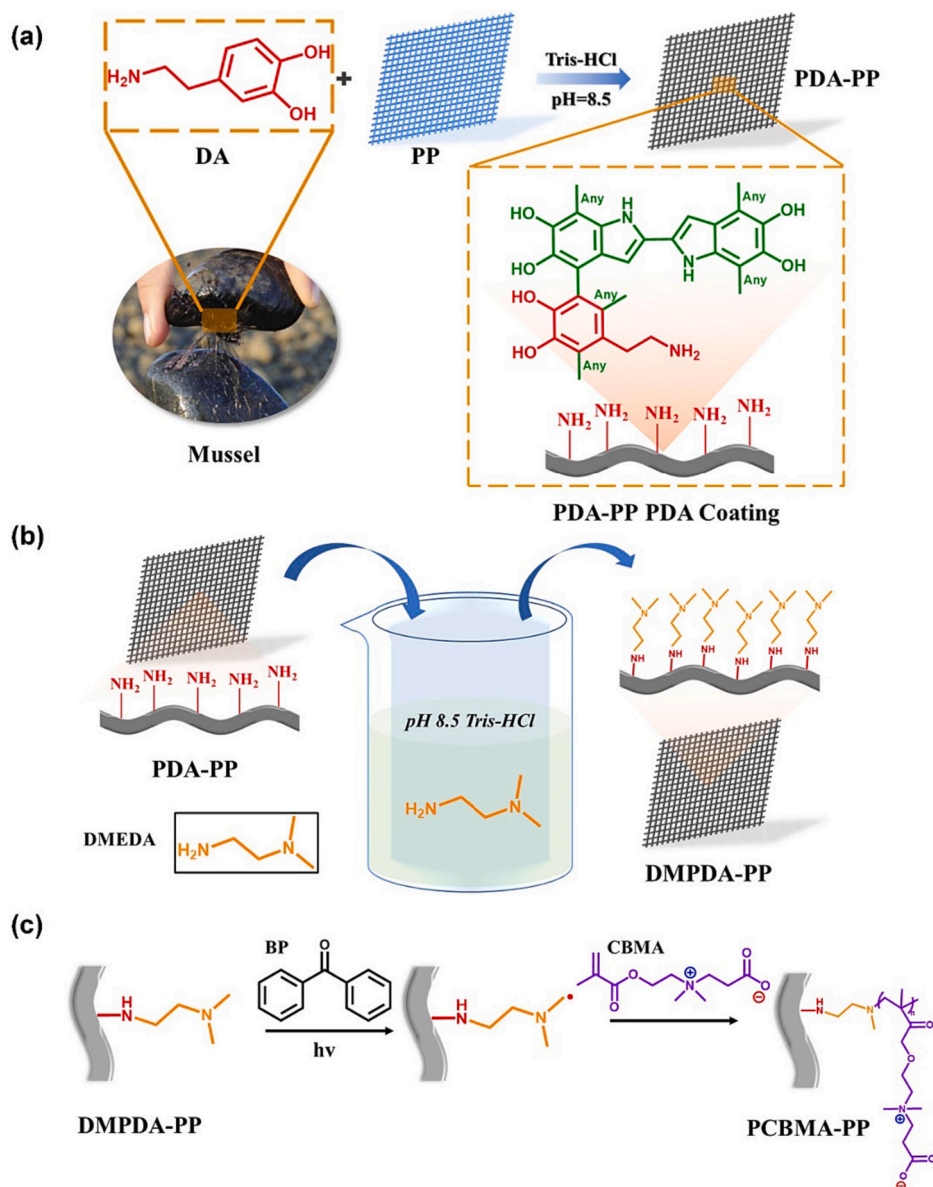


Fig. 1. The schematic showing fabrication of zwitterionic PCBMA coating on PP meshes: (a) self-aggregation of DA, (b) incorporation of DMPDA, and (c) polymerization of PCBMA on mesh surfaces.

longitudinal or transverse direction and incubated in PBS at 37 °C for 2 h. After that, each sample was stretched to a load of 16 N/cm at a speed of 100 mm/min, the constant load state was maintained for 5 s, and then the meshes returned to the initial position for 5 s. We repeated the cycle tests at least 5 times. Also, the SEM was utilized to investigate the morphological change of PCBMA-PP.

2.9. In vitro cell experiments

2.9.1. Cytotoxicity assessment

L929 mouse fibroblasts were used to assess the cytotoxicity of modified meshes. The L929 at 1×10^4 cells/well were seeded into 24-well plates with mesh samples. After 24 h of culture in a 5 % CO₂ incubator at 37 °C, the culture medium was removed, and CCK-8 solutions were subsequently added with an additional incubation of 2 h. The optical density (OD) of each group was measured at 450 nm. The cell viability was expressed as percentages relative to the OD value based on the previous work [39].

2.9.2. Anti-inflammatory activity

RAW 264.7 macrophages acted as the cell model for an *in vitro* evaluation of anti-inflammation properties. The cells at the density of 5×10^4 cells/well were seeded onto the sample and incubated for 24 h and 48 h in a humidified 5 % CO₂ atmosphere at 37 °C. For cell proliferation, the CCK-8 assay was still conducted. To observe morphologies of Raw 264.7 cells adhering to diverse meshes, all samples were rinsed with PBS buffer 3 times and fixed with 4 % paraformaldehyde overnight, followed by permeabilization with 1 % Triton and incubation with DAPI to label the nucleus. The resulting samples were investigated by the Ti-S inverted fluorescence microscope. After 24 h of culture, the supernatant was collected, and the expression of IL-6 and TNF- α was determined with the corresponding ELISA kits.

2.10. In vivo animal studies

Male Institute of Cancer Research (ICR) mice were utilized as an animal model. All animal procedures were carried out based on local animal welfare laws and approved by the Institutional Animal Ethics

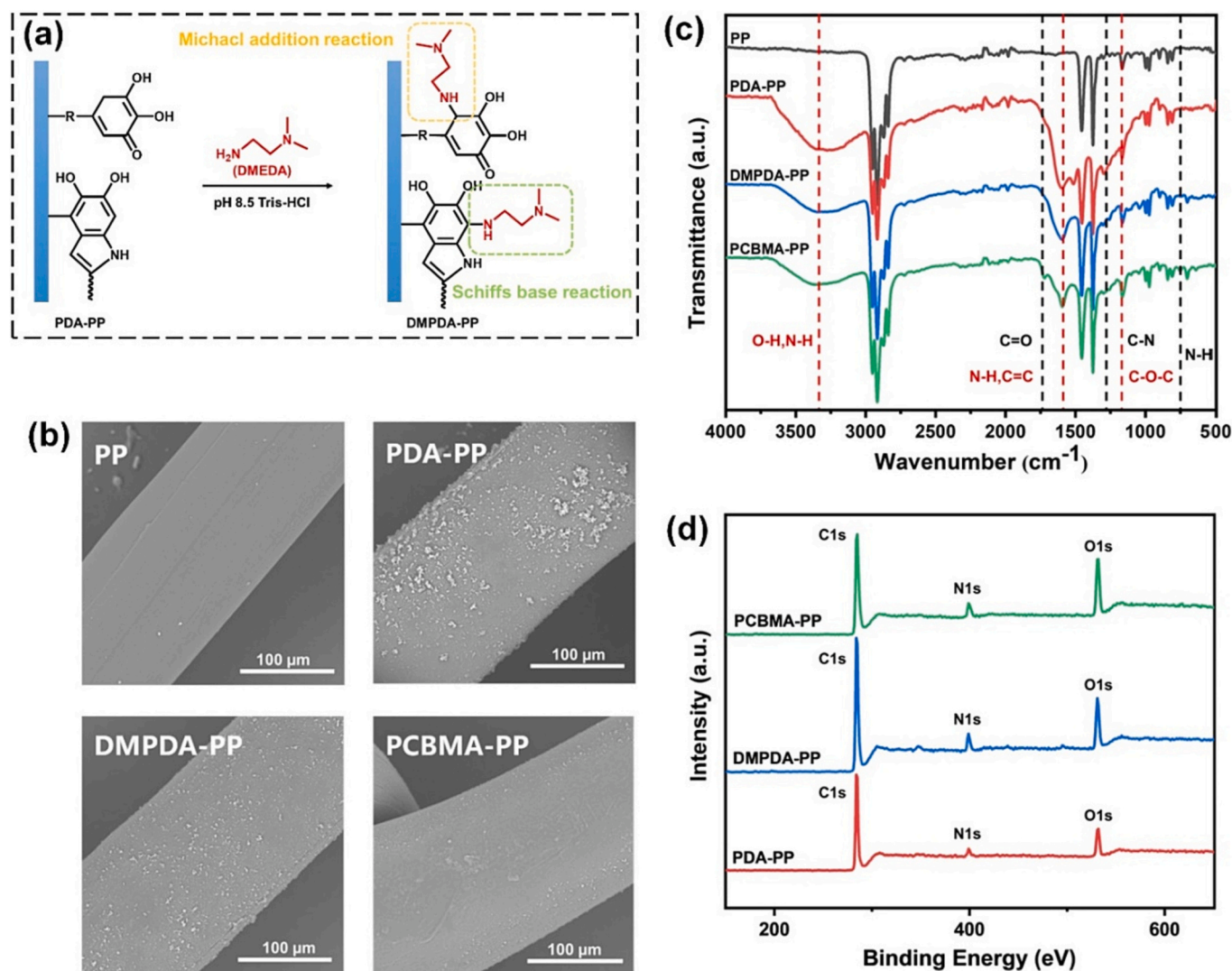


Fig. 2. (a) The reaction process of DMEDA and PDA; (b) SEM images of meshes before and after modification; (c) FTIR analyses; (d) XPS spectra.

Committee. Nine male ICR mice (eight weeks, 30–40 g) were here kept in accordance with the Guide for the Care and Use of Laboratory Animals. After anesthetization with isoflurane, the dorsal area was shaved and disinfected with 75 % ethanol. The dorsum was divided into two areas, and a 2 cm incision was made to create a subcutaneous defect for implanting a mesh sample of 1 cm × 1 cm, which was sutured with 3–0 nylon sutures. After 3, 14, and 28 days of implantation, the animals were sacrificed. All mesh samples and surrounding tissues were removed for histologic examination.

The explanted mesh/tissue samples were fixed in 10 % neutral buffered formalin, embedded in paraffin wax, and sliced into 5 μm sections for hematoxylin and eosin (H&E) and Masson's trichrome (MT) staining. The staining sections were observed using a digital pathology slide scanner (Scan Scope XT, Aperio, USA).

2.11. Statistical analyses

The statistical analyses were carried out using the one-way ANOVA test with a significant level of $p < 0.05$. Each value was reported as mean ± standard deviation. The error bars in the graphs correspond to one standard deviation. The graphs/data were marked by (*) for $p < 0.05$, (**) for $p < 0.01$ and (***) for $p < 0.001$.

3. Results and discussion

3.1. Design and fabrication of PCBMA-PP

The foreign body reaction (FBR) to implantable biomaterials can negatively affect the capability of medical devices such as hernia mesh [14,40]. Engineering coatings resisting FBR could provide hernia meshes with enhanced functionality including potentially increasing outcomes through the reduction of inflammatory response [41]. Non-specific adsorption of proteins plays a critical part in host inflammatory response and FBR initiation [18]. Inhibiting protein adsorption to mesh surfaces is a promising way to prevent inflammatory cell adhesion, aggregation, and activation, which further decreases the FBR. Hence, constructing an anti-fouling coating on hernia mesh may be a reasonable and practical approach due to its anti-adsorption effect on proteins. As shown in Fig. 1, the surface of PP mesh was activated by dopamine, and initiated by free radical polymerization of tertiary amine/photoinitiator, PCBMA coating was directly polymerized and grafted on the surface of the mesh.

3.2. Surface morphology and characterization

For PCBMA coating grafted on hernia mesh, PDA served as an intermediate layer. A stable PDA layer was spontaneously produced from

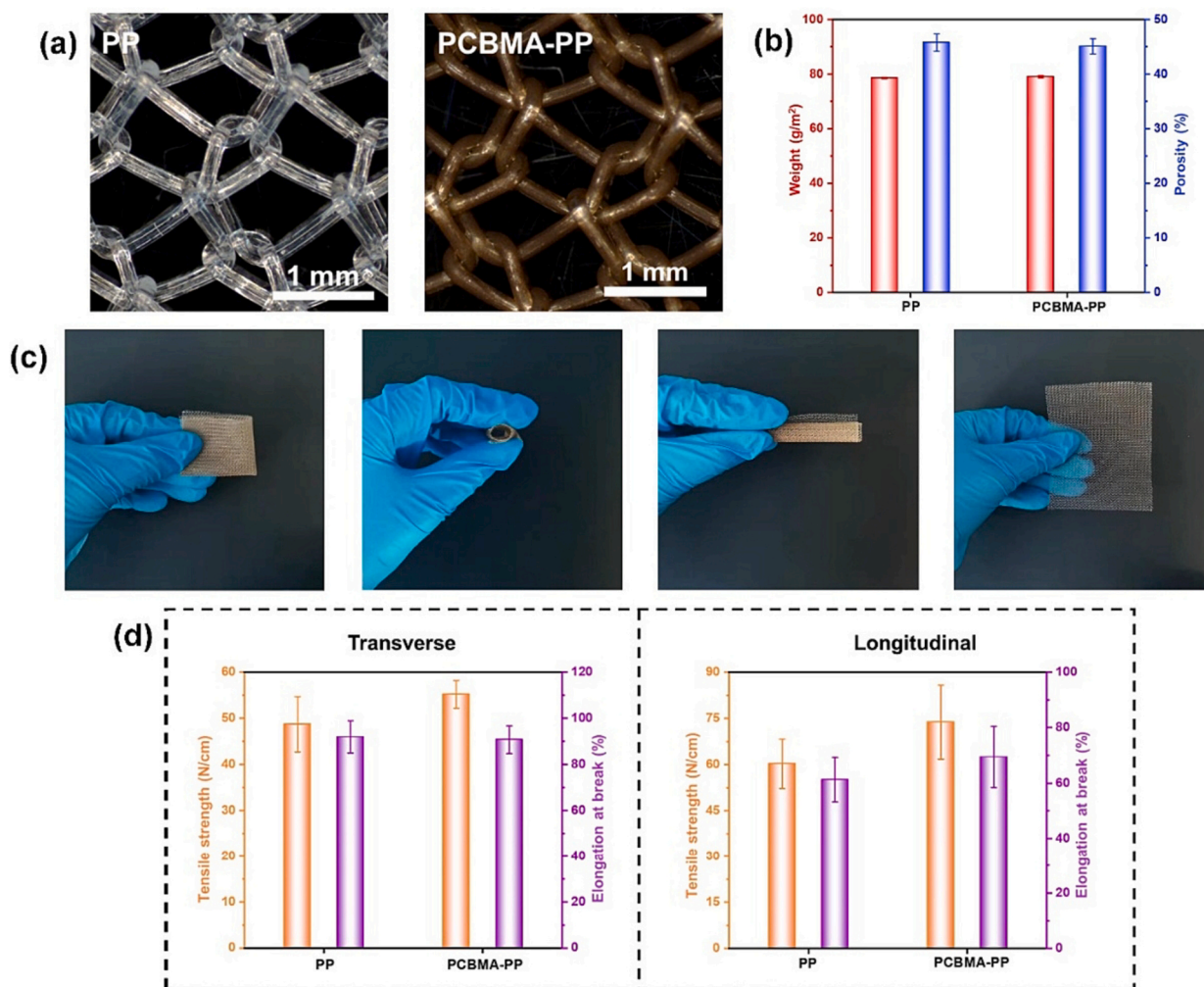


Fig. 3. (a) Microscopic observation; (b) the unit weight and porosity of mesh samples; (c) handleability of PCBMA-PP; (d) tensile mechanical properties of PP and PCBMA-PP meshes.

oxidative self-polymerization of dopamine on the surface of PP meshes in a weak alkaline condition [42], followed by the Michael addition and Schiff base reactions with DMEDA to yield tertiary amine (Fig. 2a).

Morphologies of PP meshes before and after modification were observed (Fig. 2b). From the SEM image of the initial PP mesh, it can be found that the monofilament surface is smooth. After incubation with dopamine, the *in situ* self-polymerization of dopamine occurred on the surface of PP meshes to form PDA-PP. Its corresponding SEM image displayed that particulate aggregates were deposited on the surface of PP monofilaments, suggesting that the surface-adherent PDA layer was formed. Afterward, the PDA-PP was incubated in DMEDA solution to prepare DMPDA-PP meshes. Since some of the quinones in PDA were reduced by DMEDA, the deposited particles were transformed into smaller nanoparticles with a more homogenous distribution. The SEM images of PCBMA-PP showed a relatively smoother surface, attributed to the formation of the PCBMA coating on mesh samples.

FTIR analysis was performed to further confirm that PCBMA-PP meshes were constructed successfully. In Fig. 2c, the FTIR spectra showed several characteristic peaks of different meshes between 500 and 4000 cm⁻¹. Compared with the pure PP, PDA-PP showed prominent stretching vibration peaks of the O—H and N—H bonds at 3280 cm⁻¹, resulting from phenolic hydroxyl and amino groups in PDA. The bands at 1606 cm⁻¹ and 1513 cm⁻¹ were caused by the N—H bending vibration and the C=C vibration, respectively coming from the amino group and the benzene ring structure of PDA [12]. Moreover, the C—N bond

stretching vibration peak at 1296 cm⁻¹ also proves the existence of PDA on the surface of PDA-PP. In the spectrum of DMPDA-PP, the C=C vibration peak at 1513 cm⁻¹ disappears due to the PDA layer covered by DMEDA. A new peak appears around 700 cm⁻¹ assigned to the N—H bending vibration of DMEDA, confirming the successful construction of DMPDA-PP. In comparison, a noticeable C=O stretching vibration peak around 1723 cm⁻¹ was observed in the spectrum of PCBMA-PP, resulting from carboxyl groups of PCBMA [43]. The above results indicated the sequential deposition of PDA, DMEDA, and PCBMA on the surface of PP meshes. The facile method can be applied to fabricate PCBMA-PP, effectively.

Surface chemical compositions of different meshes were examined by XPS measurements. The survey-scan spectra were recorded for PDA-PP, DMPDA-PP, and PCBMA-PP in Fig. 2d. Apparently, all the mesh samples exhibited the characteristic peaks of N1s and O1s. Compare to PDA-PP with the N element content of 5.13 %, that of DMPDA-PP increased to 7.49 %. No doubt, PDA contains many O elements, while DMEDA only contains C and N elements. However, the peak of O was detectable on DMPDA-PP, because some of the PDA was still exposed on the mesh surface, while the loading of DMEDA decreased the element ratios of O/N and C/N. PCBMA-PP should contain more O elements than DMPDA-PP. As illustrated in Table S1, the O content on the mesh surface was increased from 13.30 % to 17.76 % after grafting PCBMA. Also, the O/N ratio rose markedly. The high-resolution spectra of C1s (Fig. S1) displayed three peaks around 284.6 eV, 285.8 eV, and 287.5 eV, mainly

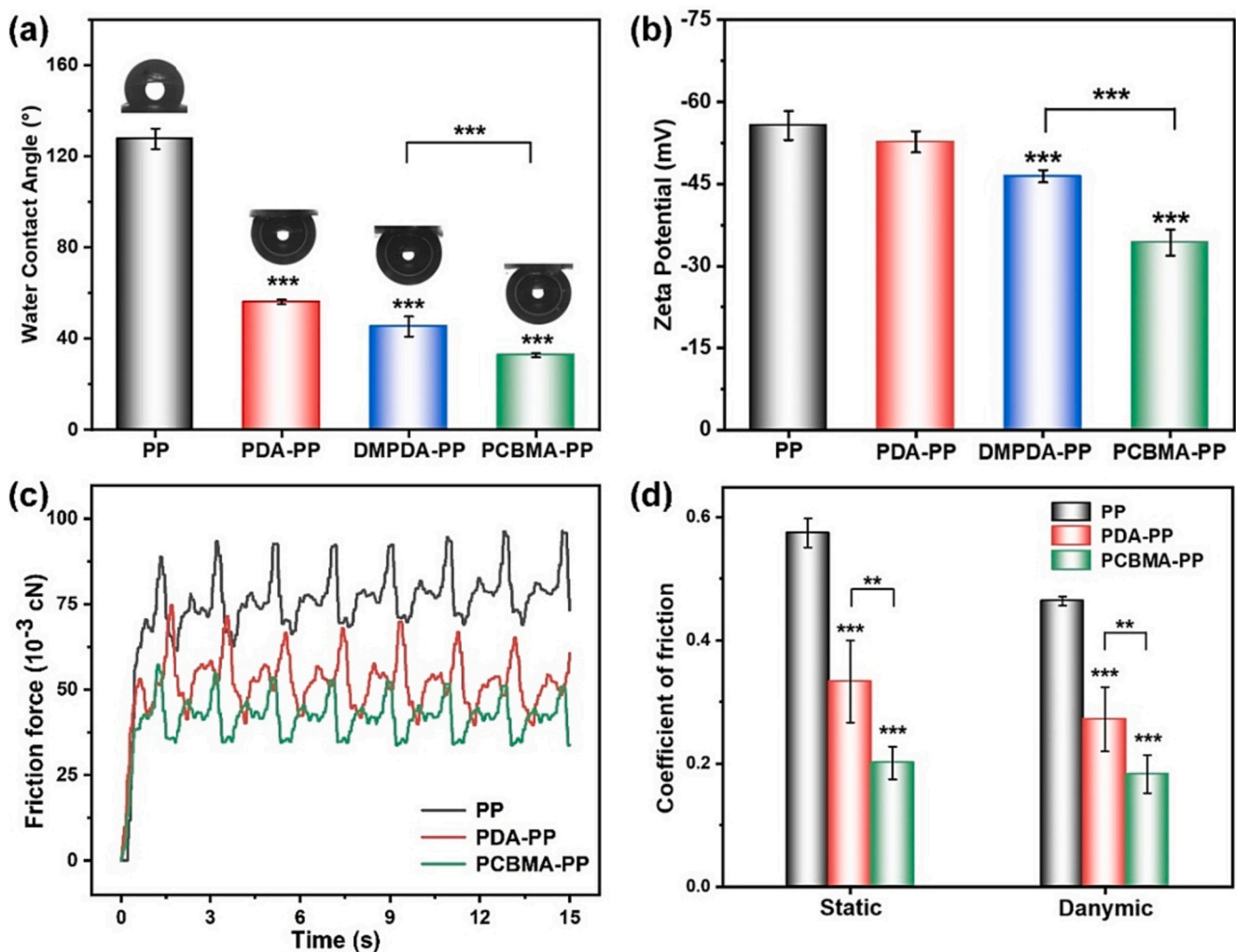


Fig. 4. (a) The WCA measurement; (b) ζ -potential; (c) friction force; (d) coefficient friction of diverse mesh samples.

corresponding to C—C, C-N/C-O, and C=O. For DMPDA-PP, the structural peak at 285.8 eV was stronger than that of PDA-PP. By contrast, PCBMA-PP showed a higher structural peak of C=O due to the presence of carboxyl groups on the mesh. The fitted XPS spectra of N1s for PDA-PP had three structural peaks, which were characteristic of aromatic N (398.1 eV), C-NH (399.5 eV), and C-NH₃⁺ (401.2 eV). The aromatic N is only present on PDA-PP, consistent with FTIR analyses. The production of C-NH₃⁺ might be due to the spontaneous proton transfer from the catechol to the amine [44]. The N1s spectrum of PCBMA-PP exhibited a higher peak located at 401.2 eV, attributed to the pendant group of PCBMA. In conclusion, XPS analyses of PDA-PP, DMEDA-PP, and PCBMA-PP confirmed the successful loading of PDA, DMEDA, and PCBMA layers, correspondingly.

As shown in Fig. 3a, visually, the pristine PP mesh was white and transparent. After zwitterionic modification, the obtained PCBMA-PP mesh retained the warp-knitted porous structure, but the color turned yellow-brown due to the oxidative self-aggregation of DA on the mesh surface. The PDA particles were black-brown, resulting from dopamine quinone and 5, 6-dihydroxyindole. During the DMEDA grafting process, some of the quinones were increased, making black-brown lighter. Additionally, PCBMA is a transparent polymer, and its coverage on the monofilaments also lightened the color of PDA. Finally, the PCBMA-PP showed a yellow-brown surface.

A standard PP mesh (78.6 ± 0.2 g/cm², 45.8 ± 1.6 %) here was utilized as base materials for modification to reduce FBR. As shown in Fig. 3b, there was no significant difference in the unit weight and

porosity between the PP and the PCBMA-PP (79.0 ± 0.4 g/cm², 45.1 ± 1.4 %). Moreover, the filament diameter of PP is similar to that of PCBMA-PP (Table S2), indicating that the modified coating did not notably alter the monofilament diameter of mesh samples.

3.3. Tensile mechanical property

As shown in Fig. 3c, the PCBMA-PP mesh had good handling properties. To further explore whether the PCBMA coating would affect the mechanical strength, we compared PP and PCBMA-PP in uniaxial tensile testing. As listed in Table S3, with the incorporation of PCBMA coatings, the transverse and longitudinal tensile strengths of mesh samples increased by 13.3 % and 22.5 %, respectively. There was no notable difference in horizontal elongation between PP and PCBMA-PP meshes, while the longitudinal elongation of PCBMA-PP increased by 8.21 %. Although PCBMA coatings could impact the mechanical properties, no significant change in tensile strength and elongation at break was observed (Fig. 3d). Accordingly, mesh samples could retain the essential mechanical properties after the incorporation of PCBMA.

3.4. Wettability and ζ -potential

To analyze the change in wettability for modified meshes, a WCA assessment was performed. From Fig. 4a, it could be seen that the hydrophilicity of mesh samples was largely improved by the deposition of the PDA layer (WCA $\approx 56^\circ$). After depositing DMEDA, the WCA of

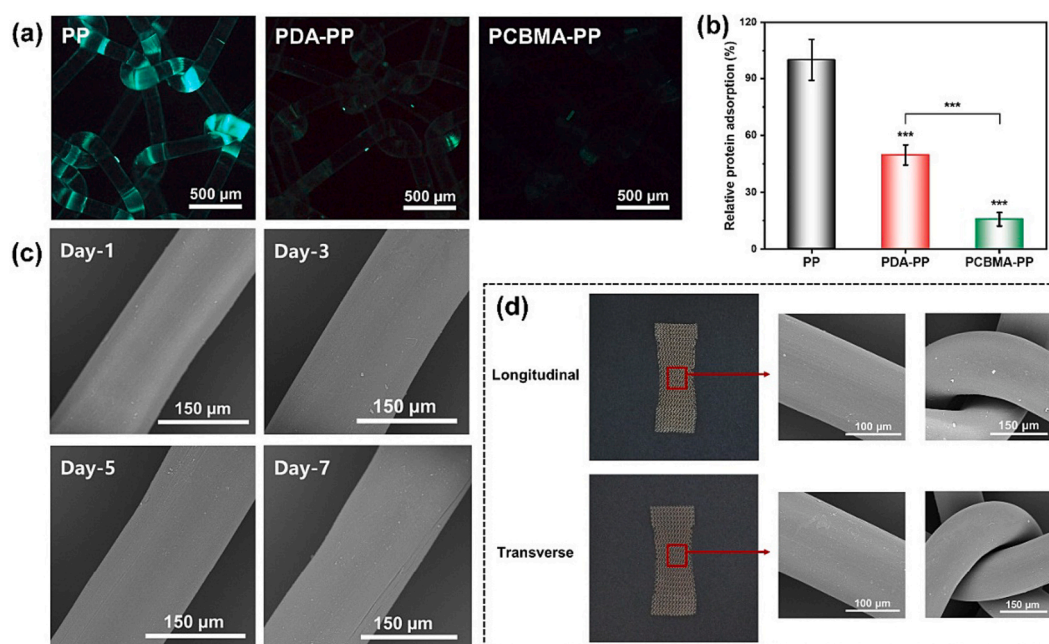


Fig. 5. (a) Fluorescence images of FITC-BSA adsorbed on different mesh surfaces; (b) relative protein adsorption; (c) SEM images of PCBMA-PP at different degradation times; (d) macroscopic and microscopic observation on PCBMA-PP during dynamic degradation.

meshes decreased to 45° , to which the increased amino group contributed. In contrast, PCBMA-PP had lower WCA (32.8° , $p < 0.001$), suggesting better hydrophilicity. As shown in Fig. 1c, PCBMA-PP had moieties bearing both cationic quaternary ammonium and anionic carboxylate groups, being able to combine with water molecules through electrostatic interaction and hydrogen bonding, thus endowing the PCBMA-PP mesh with superior hydrophilicity [45].

With spectroscopic ellipsometry, the measured thickness of PCBMA coating was only $1.12 \pm 0.27 \mu\text{m}$. Further, ζ -potential measurement was applied to assess the potential changes of mesh surfaces before and after modification. As observed in Fig. 4b, the initial PP mesh exhibited a negative surface charge, about -55.7 mV , in a negatively charged polystyrene bead solution system. The PDA coating had no significant impact on the surface charge of mesh samples ($p > 0.05$). With DMPDA incorporated, the surface potential of modified meshes rose to -46.4 mV , largely. This is attributed to the presence of protonated amino groups on DMPDA-PP. Since zwitterionic polymer PCBMA has an equal amount of positive and negative charges, the incorporation of PCBMA coatings could neutralize part of surface charges. Particularly, the zeta potential of PCBMA-PP increased to -34.3 mV . Negatively charged proteins and cell membranes are ubiquitous in human body. Theoretically, higher zeta-potential enhances charge repulsion between proteins and mesh surface, contributing to the decrease of protein adsorption. While the coated meshes still showed a negative surface charge. Additionally, in aqueous solutions, surface charges are shielded by hydrating water. Presumably, hydrophilic surfaces are generally reported to adsorb fewer proteins than hydrophobic ones due to a greater number of possible adsorption-promoting interactions.

3.5. Hydration lubrication

The hydration lubrication was closely related to the interfacial interaction between mesh materials and proteins after implantation. Here, tribological testing was conducted to evaluate the lubrication properties of various meshes (PP, PDA-PP, and PCBMA-PP). If the surface of PP monofilaments was grafted with the PCBMA coating, PCBMA would absorb water and form a hydration layer on mesh surfaces. So, all the mesh samples were equilibrated for 2 h in the same aqueous solution. The changes in sliding friction and friction coefficient of meshes

before and after modification were displayed in Fig. 4c and d. For PCBMA-PP meshes, the surface friction coefficients decreased remarkably as a result of hydration lubrication [46,47]. Compared with PP meshes, static and dynamic friction coefficients of PCBMA-PP were decreased by 65.0 % and 60.8 %, respectively. In brief, the PCBMA layer endowed PCBMA-PP with excellent lubricating properties.

3.6. Protein adsorption

It is known that proteins consist of >20 amino acids, which have different physicochemical properties due to their distinct sequences. Most proteins in body fluids are negatively charged hydrophobic structures, so the surface properties of implant materials can affect protein adsorption. When biomedical devices were implanted into the human body, they will contact various proteins and cells in the blood. Particularly, the adsorption of non-specific proteins is described as the first step of interaction between device materials and the human body [48]. Accordingly, in the design of an effective anti-inflammatory strategy, we should avoid non-specific protein adsorption in the feature development of hernia mesh. As the most abundant vascular protein, albumin dominates the initial interaction with mesh surfaces to cause inflammation. Here, we chose commonly used BSA as the model protein. In Fig. 5a, it is visible that PP adsorbed most proteins with the biggest fluorescence areas. Fewer proteins were observed on the surface of PDA-PP. The least proteins adsorbed on PCBMA-PP showed the smallest fluorescence area. The quantitative evaluation utilizing a BCA kit also confirmed these results. The PDA layer decreased protein adsorption on mesh surfaces largely, due to better hydrophilicity than PP. PCBMA-PP had a statistically significant reduction in protein adsorption compared with the others. It may be that the hydration ability of PCBMA formed a strong water barrier to make the mesh surface slippery, thus improving anti-fouling properties and reducing protein adsorption [49]. Additionally, conformational changes of adsorbed proteins may influence cell adhesion and the subsequent immune responses [16]. As a zwitterion, PCBMA is a neutral compound [50], which has very weak interactions with BSA molecules and won't elicit changes in the structural conformation of proteins. Thus, the PCBMA-PP with minimal protein adsorption may reduce immune recognition.

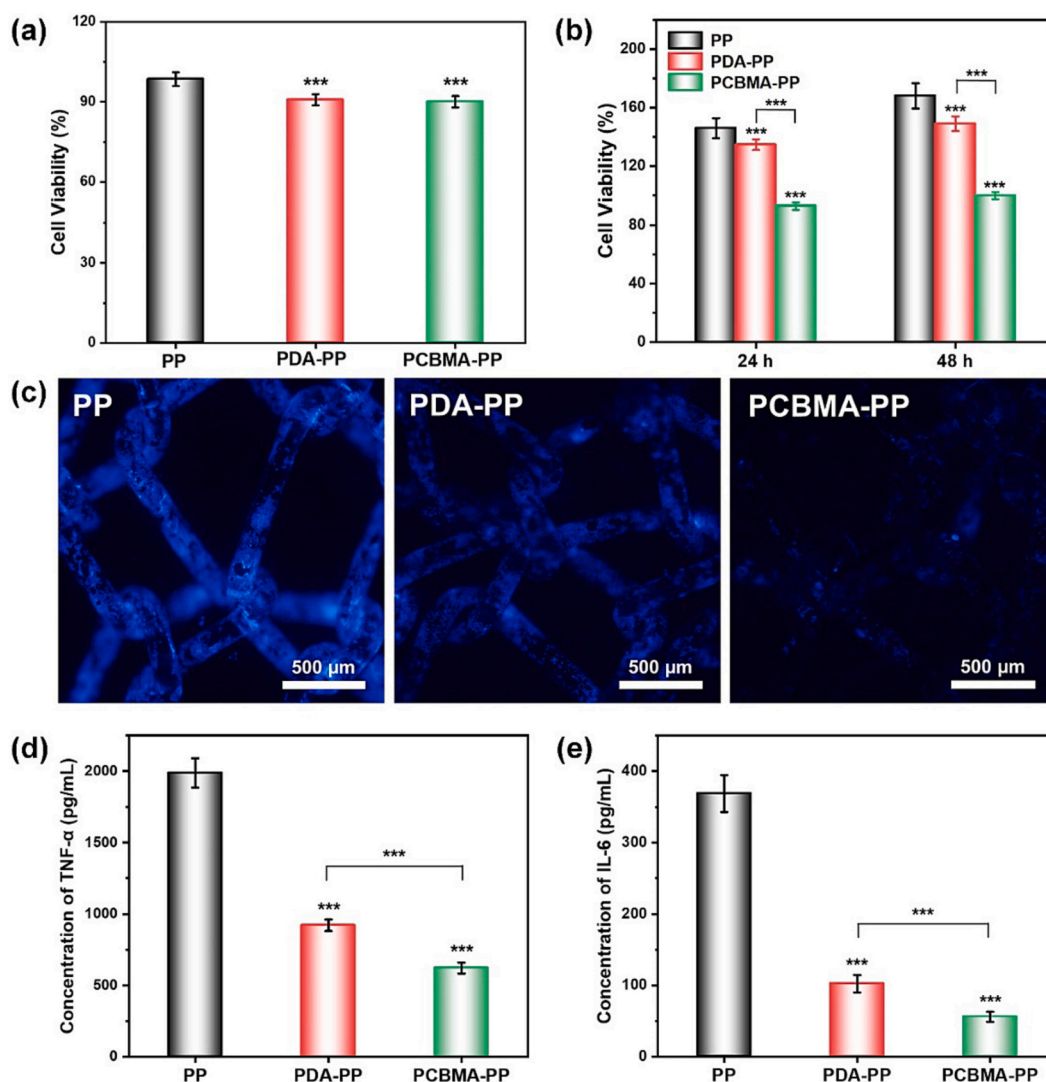


Fig. 6. (a) Cell viability of L-929 fibroblasts cultured with different meshes after 1 day; (b) cell viability of RAW 264.7 macrophages after incubation of 1 and 3 days; (c) fluorescence images with DAPI staining; (d) TNF- α expression; (e) IL-6 expression in RAW 264.7 macrophages after 1 day of culture.

3.7. Stability of PCBMA-PP

To examine the stability of the PCBMA layer, a series of studies were performed to evaluate the degradation of PCBMA-PP after incubation for 7 days in PBS puffer (pH = 7.4). The assessment was based on the persistence of PCBMA coating at different times. In Fig. 5c, PCBMA-PP showed stable surface morphologies after degradation of 7 days. Also, the mesh surface was maintained smooth without notable changes. In conclusion, the deposition coating persisted strongly and remained excellently stable.

For mesh materials, the mechanical property is a crucial parameter for their successful implant, which should provide sufficient reinforcement to promote abdominal wall restoration. If the abdominal cavity is assumed to be a thin hollow sphere (3–8 cm in diameter), the rupture forces are around 4–16 N/cm [51]. To further assess the stability of deposition coating, PCBMA-PP was first soaked in PBS solution and was sufficiently hydrated in a simulated environment, followed by cyclic tensile testing under a loading of 16 N/cm. From Fig. 5d, it can be seen that whether the meshes were stretched horizontally or vertically, the PCBMA coating remained intact on the mesh. There is no coating failure and spallation for each monofilament, and the coating surface is still uniform and smooth without visible cracking.

3.8. In vitro biological evaluation

Prosthetic meshes should be non-toxic to cell growth [3]. Here, L-929 fibroblasts were utilized to evaluate the cytocompatibility of each mesh. The results of CCK-8 assays were illustrated in Fig. 6a. After 24 h in co-culture, the corresponding cell viability was 98.49 %, 90.76 %, and 90.03 % for PP, PDA-PP, and PCBMA-PP, suggesting good cell compatibility.

Also, the CCK-8 detection was run to examine the viability of RAW 264.7 macrophages. After 24 h and 48 h, the viability of cells cultured with PP was 145.9 % and 168.0 %, respectively. It can be inferred that PP promotes cell adhesion and proliferation of RAW 264.7. In comparison, PDA-PP showed a significantly reduced cell viability, which was correspondingly 134.6 % and 148.9 %. This is owing to better hydrophilicity and anti-protein adsorption property of PDA-PP than those of PP [52]. In apparent, PCBMA-PP exhibited the lowest cell viability, which decreased to 92.8 % and 99.0 % at 24 h and 48 h, indicating the ability to inhibit macrophage survival, adhesion, and proliferation.

The morphology of macrophages cultured with different meshes was investigated under fluorescence microscopy after seeding for 24 h. In Fig. 6c, numerous macrophages adhered to the monofilament of PP. The number of cells on the surface of PDA-PP and PCBMA-PP decreased, while fewer were on the latter. The reason for this phenomenon is that

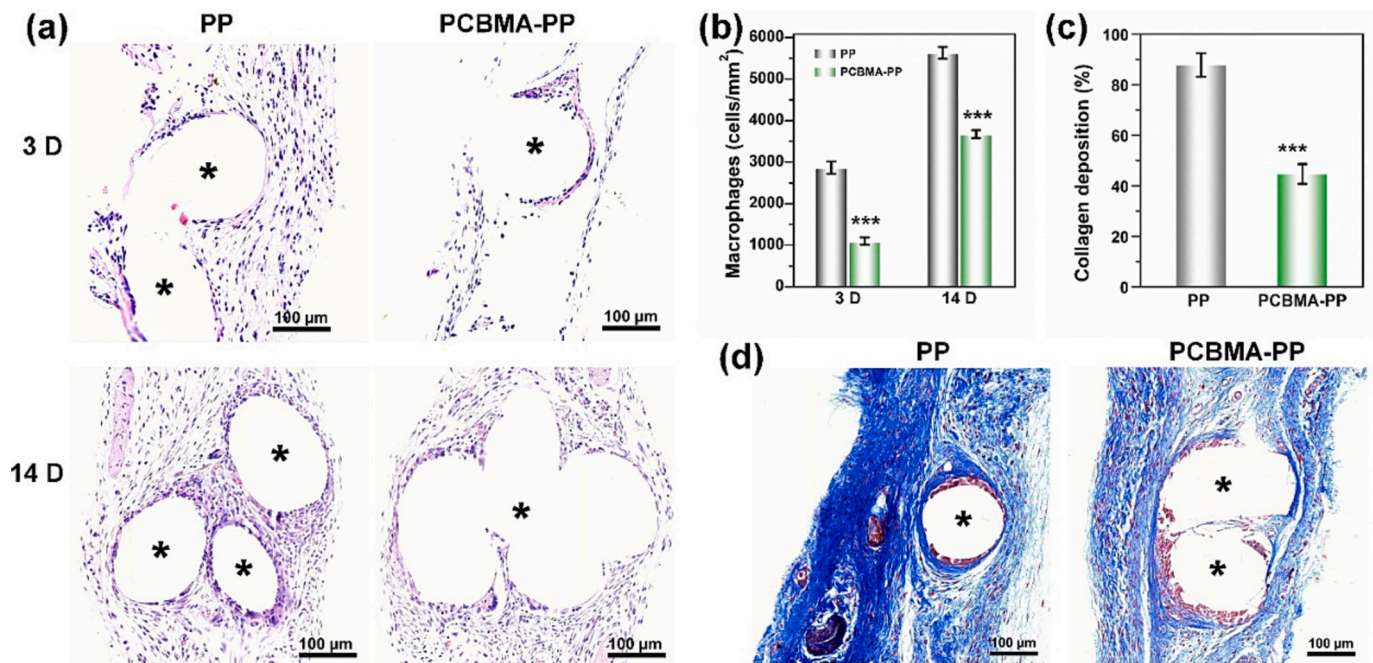


Fig. 7. (a) Micrographs of HE staining; (b) macrophages at 3 and 14 days; (c) collagen deposition at 28 days; (d) MT staining at 28 days in tissue slices.

the improved hydrophilicity can reduce non-covalent forces between proteins/cells and mesh materials, thus effectively minimizing protein adsorption and cell adhesion. However, there are fine particles on the mesh surface after depositing PDA. The less smooth the surface of meshes, the easier it is for cells to spread. Thereby, PDA-PP exhibited a limited effect on preventing cell adhesion. After combination with water molecules, the PCBMA coating can form a hydration layer on mesh surfaces, which has super lubricating properties to inhibit macrophage adhesion and proliferation. Briefly, the PCBMA layer makes the probability of meshes with super anti-inflammatory properties.

Macrophage activation or polarization plays an essential part in inflammation progress. When macrophages are activated, they can be polarized into a proinflammatory phenotype, which can secrete proinflammatory cytokines to aggravate inflammation [17]. Herein, the expression of TNF- α and IL-6, two key proinflammatory cytokines, were examined using ELISA kits. As shown in Fig. 6d and e, after culturing 24 h, PP featured the highest level of TNF- α and IL-6 at 1986.9 and 968.9 pg/mL, suggesting that macrophages were activated to be proinflammatory by the initial PP mesh. PDA-PP can elicit a relatively lower inflammatory response, as >50 % less TNF- α (920.3 pg/mL) and IL-6 (103.2 pg/mL) were released than that of PP ($p < 0.05$), resulting from its anti-protein adsorption and antioxidant properties [18]. Both modified meshes exhibited anti-inflammatory effects, but PCBMA-PP was found to be more effective in mitigating inflammation. PCBMA-PP had significantly decreased levels of TNF- α (620.67 pg/mL) and IL-6 (57.5 pg/mL) compared to PP and PDA-PP ($p < 0.05$), attributed to its better hydration lubricating ability and stronger protein resistance.

3.9. In vivo assessments

Anti-inflammatory performance *in vivo* of modified meshes was studied in a subcutaneous mouse model. Related assessments of the implants were performed from three aspects: acute inflammation (3D), chronic inflammation (14D), and collagen deposition (28D).

The micrographs of HE staining in tissue slices from PP and PCBMA-PP implants are shown in Fig. 7a. After implantation of 3 and 14 days, the HE staining images represented the accumulation of inflammatory cells in the surrounding tissue around monofilaments. Compared to PP, the inflammatory cells surrounding PCBMA-PP were fewer and more

uniformly distributed. Due to the existence of implants, macrophages accumulate and fuse to generate foreign body giant cells at the tissue-implant interface over time. To quantify the density of macrophages, the number of cells distributed around each sample was counted using ImageJ software. As shown in Fig. 7b, PCBMA-PP recruited approximately 1094 and 3674 cells/mm² of macrophages after 3 and 14 days, significantly lower than that of PP (2862 and 5631 cells/mm², respectively), suggesting the alleviation of local inflammation. The incorporation of PCBMA coating reduced the inflammatory response, as the presence of zwitterionic layer with anti-protein adsorption property and cytocompatibility decreased the stimulation of PP to the adjacent tissue.

Collagen deposition is the key to tissue regeneration and function to be restored after hernia repair [5,53]. MT staining was conducted to analyze the collagen deposited around each sample at 28 days of implantation. The results were shown in Fig. 7c and d. The micrographs showed that normal tissue integration was investigated for PCBMA-PP due to the permeability of preserved pores, while denser collagen fibers stained blue are distributed around the monofilaments of PP than PCBMA-PP. Fig. 7c showed the relative collagen amount of each group. Significantly excessive collagen deposition was found in PP (87.8 %) than in PCBMA-PP (44.7 %). Excessive collagen deposition is likely to cause fibrosis, which has a straight relation with inflammation [54,55]. It means that severe inflammatory responses can promote dense fibrous intergradation. The above results revealed that PCBMA-PP could inhibit overexpression of collagen deposition and exhibit superior tissue compatibility.

4. Conclusions

In this study, we developed a facile strategy to construct an anti-fouling coating on PP mesh with improved anti-protein adsorption and anti-inflammatory properties. The resulting mesh was proven to exhibit notably increased hydrophilicity, hydration lubrication, and excellent stability as incorporating the zwitterionic PCBMA layer. Compared with the unmodified PP mesh, PCBMA-PP had significantly stronger anti-protein adsorption properties, the increased ability to prevent inflammation and collagen deposition, and better tissue compatibility, thus showing great potential for efficient hernia repair.

CRediT authorship contribution statement

Ying Mao: Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Qian Wang:** Formal analysis, Investigation, Methodology, Writing – review & editing. **Huiru Zhang:** Formal analysis, Writing – review & editing. **Yan Li:** Resources, Supervision, Writing – review & editing. **Lu Wang:** Resources, Supervision.

Declaration of competing interest

No conflict of interests.

Data availability

Data will be made available from the corresponding authors on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bioadv.2024.213769>.

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